

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (CE/IT)

December 2011

CE 210 Microprocessor & Computer Architecture

Date: 10.12.2011, Saturday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

## Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

## SECTION - I

Q - 1 (a) Do as direct.

[07]

- 1) How many bits are stored by a  $256 \times 4$  memory chip? Can this chip be specified as 128 byte memory?
- 2) How many address lines necessary on the chip of 2048(2K) byte memory?
- 3) specify the crystal frequency required for an 8085 system to operate at 1.1 MHz.
- 4) Explain the need to demultiplex the bus  $AD_7-AD_0$ .
- 5) Specify the 8085 signals that are used to enable an input port.
- 6) What are the control signals necessary in the memory mapped I/O?
- 7) What is the function of ALE and IO/M signal of 8085?

Q - 2 (a) Explain the architecture of the 8085 microprocessor with its flag register.

[07]

(b) Why is the program counter and stack pointer 16 bit register in 8085?

[03]

(c) Explain the following instructions.

[04]

1.SHLD 2000H 2.PCHL 3.XTHL 4.DAA

## OR

Q - 2 (a) Explain 8085 interrupts and externally initiated signals.

[07]

(b) Write a note on memory.

[03]

(c) Write a program to find 2's complement of given data which is stored at 2200H memory location.

[04]

Q - 3 (a) Write a program to swap two hexadecimal data which is stored at 2000H and 2200H memory location.

[07]

(b) Write a note on push and pop instruction with suitable example.

[07]

## OR

Q - 3 (a) What is subroutine? Give the detail of 8085 subroutine instruction with suitable example.

[07]

- (b) List out logical, rotate and compare instruction in 8085. explain in detail. [07]

**SECTION – II**

- Q - 4 (a) Describe block diagram of 8255 ICs with its internal block. [07]

- Q - 5 (a) Draw and explain block diagram of 8259 programmable interrupt controller with its command word. [07]

- (b) Write a note on Pentium III microprocessor with suitable example. [07]

**OR**

- Q - 5 (a) What is synchronous and asynchronous transmission? Describe 8251 USART ICs with its different sections. [07]

- (b) Draw and explain 8086 microprocessor with its flag register. [07]

- Q - 6 (a) Describe Pentium pro processor with its internal structure. [07]

- (b) Describe control register structure of 80486 processor. [07]

**OR**

- Q - 6 (a) Write a note on internal structure of 80386 processor. [07]

- (b) List out various addressing mode of 80286 processor in detail. [07]

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